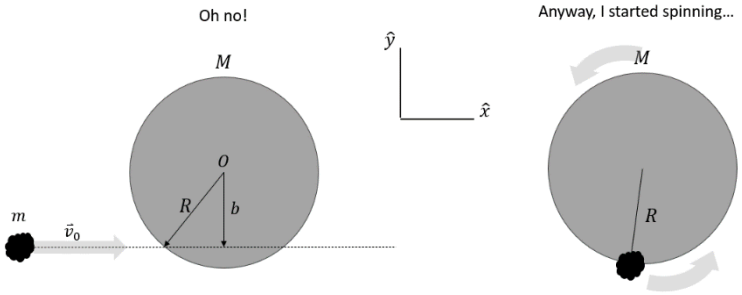
PHY-275 Fall 2026 Mid-term

Name: _____ COE-email: _____

1 credit for each task, and (sometimes) extra tokens with each successful step.

Q1: Angular Momentum Conservation in an Inelastic Impact



A uniform solid disk of mass M and radius R lies in a horizontal plane and is mounted on a frictionless vertical spindle passing through its center O. The disk is initially at rest. A small sticky blob of putty of mass m slides along the horizontal table toward the disk at a constant velocity $\vec{v}_0$. The line of motion of the putty lies at an impact parameter b ($0 < b < R$) relative to the disk's center O. The putty strikes the disk and sticks to its outer edge at distance R from the center O.

(a) Derive the total moment of inertia I_{total} of the system after the collision. Using angular momentum conservation, derive an expression for the final angular velocity ω_f of the disk-putty system in terms of m M $\vec{v}_0$ b and R. Hint: moment of inertia of uniform disk is: $I_{\text{disk}} = \frac{1}{2}MR^2$. [1 token]

$$I_{\text{total}} = I_{\text{disk}} + I_{\text{putty}} = \frac{1}{2}MR^2 + mR^2 = \left(\frac{1}{2}M + m\right)R^2 = \frac{M+2m}{2}R^2$$

$$mv_0 b = \left(\frac{1}{2}M + m\right)R^2\omega_f \implies \omega_f = \frac{mv_0 b}{\left(\frac{1}{2}M + m\right)R^2} = \frac{2mv_0 b}{(M+2m)R^2}$$

(b) Calculate the ratio of the final mechanical energy E_f to the initial mechanical energy E_i of the system. [1 token]

$$E_i = \frac{1}{2}mv_0^2 \qquad E_f = \frac{1}{2}I_{\text{total}}\omega_f^2 = \frac{1}{2}\frac{L_f^2}{I_{\text{total}}} = \frac{1}{2}\frac{(mv_0 b)^2}{\left(\frac{1}{2}M + m\right)R^2}$$

$$\frac{E_f}{E_i} = \frac{\frac{1}{2}\frac{m^2v_0^2b^2}{\left(\frac{1}{2}M+m\right)R^2}}{\frac{1}{2}mv_0^2} = \frac{mb^2}{\left(\frac{1}{2}M + m\right)R^2} = \frac{2mb^2}{(M+2m)R^2} = \left(\frac{2m}{M+2m}\right)\left(\frac{b}{R}\right)^2$$

Q2: Show me how Euler's formula is derived via Taylor series expansion, all necessary formula is provided as supplements.

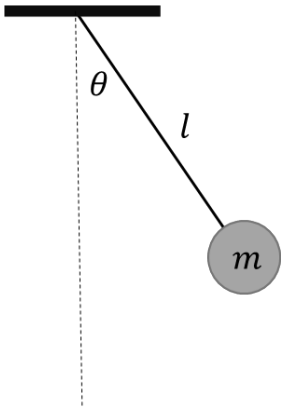
$$e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!} = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \frac{u^4}{4!} + \frac{u^5}{5!} + \frac{u^6}{6!} + \frac{u^7}{7!} + \dots$$

$$\begin{aligned} e^{i\theta} &= 1 + (i\theta) + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \frac{(i\theta)^6}{6!} + \frac{(i\theta)^7}{7!} + \dots \\ &= 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i\frac{\theta^5}{5!} - \frac{\theta^6}{6!} - i\frac{\theta^7}{7!} + \dots \end{aligned}$$

$$e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots\right)$$

$$\cos\theta = \sum_{k=0}^{\infty} (-1)^k \frac{\theta^{2k}}{(2k)!} = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots, \quad \sin\theta = \sum_{k=0}^{\infty} (-1)^k \frac{\theta^{2k+1}}{(2k+1)!} = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$$

Q3: With small angle approximation, a single pendulum is a harmonic oscillator:



You have point mass m, rigid rod of length l, gravity field g, angular displacement θ. Initial condition: @ $t = 0,\ \theta = \theta_0,\ \dot{\theta} = 0$

(a) Derive the equation of motion and apply the small-angle approximation.
(b) Guess solution as $e^{\lambda t}$ find the specific solution based on initial condition. (1 token)
(c) Guess solution as $\cos \omega t$ and find the specific solution based on initial condition. (1 token)
(d) Find the relationship between λ and ω. (1 token)

(a) $I\ddot{\theta} = \tau \quad \ddot{\theta} + \frac{g}{l}\sin\theta = 0$ so $\ddot{\theta} + \frac{g}{l}\theta = 0$ small angle approx.

(b) $\left(\lambda^2 + \frac{g}{l}\right)e^{\lambda t} = 0$

$$\lambda^2 = -\frac{g}{l} \quad \lambda = \pm i\sqrt{\frac{g}{l}}$$

$$\theta(t) = c_1 e^{i\sqrt{g/l}\,t} + c_2 e^{-i\sqrt{g/l}\,t}$$, given i.c.:

$$\theta(0) = c_1 + c_2 = \theta_0$$
$$\dot{\theta}(0) = i\sqrt{\frac{g}{l}}(c_1 - c_2) = 0 \implies c_1 = c_2$$

$$\theta(t) = \frac{\theta_0}{2}e^{i\sqrt{g/l}\,t} + \frac{\theta_0}{2}e^{-i\sqrt{g/l}\,t}$$

(c) $\theta(t) = \cos(\omega t)$

$$\ddot{\theta}(t) = -\omega^2\cos(\omega t)$$

$\left(-\omega^2 + \frac{g}{l}\right)\cos(\omega t) = 0$ that's $\omega = \pm\sqrt{\frac{g}{l}}$

$\theta(t) = Acos(\omega t) + Bcos(\omega t)$ given i.c.:

$\theta(0) = A + B = \theta_0$

$\dot{\theta}(0) = -\omega A + \omega B = 0$

$$\theta(t) = \theta_0 \cos(\omega t) = \theta_0 \cos\left(\sqrt{\frac{g}{l}}\, t\right)$$

(d) $\lambda^2 = -\omega^2 = -\frac{g}{l}$ or $\lambda = \pm i\omega$

Q4 Write down the 2nd order ODE $\ddot{x} + K_1\dot{x} + K_2 x = 0$ that has eigenvalues $\lambda = C \; and \; \lambda = D$. (Hint: to find $K_1 \; and \; K_2$ with given eigenvalues.) If the solution is $x(t) = Ae^{Ct} + Be^{Dt}$, what are the initial conditions?

Characteristic equation is: $\lambda^2 + K_1\lambda + K_2 = 0$, also:

$$(\lambda - C)(\lambda - D) = \lambda^2 - (C + D)\lambda + CD = 0$$, so:

$K_1 = -(C + D), \quad K_2 = CD$, that's $\ddot{x} - (C + D)\dot{x} + CD\, x = 0$

$x(0) = Ae^{C(0)} + Be^{D(0)} = A + B$

$\dot{x}(t) = ACe^{Ct} + BDe^{Dt} \implies \dot{x}(0) = CA + DB$



Q5 Multi-choice: SO-101 is a 6 DOF robotic arm with the list of joints and ranges of turning angles shown on the left. When solving the forward kinematics using DH parameter, what is the **size** of the overall homogeneous (the final single) transformation matrix relating the gripper position to the base frame?

(A) 6x4
(B) 4x6
(C) 4x4
(D) 3x4

Q6 The potential energy of a certain particle is $U = Axy^2 + Bsin(Cz)$, where A, B, and C are constants. What is the corresponding force?

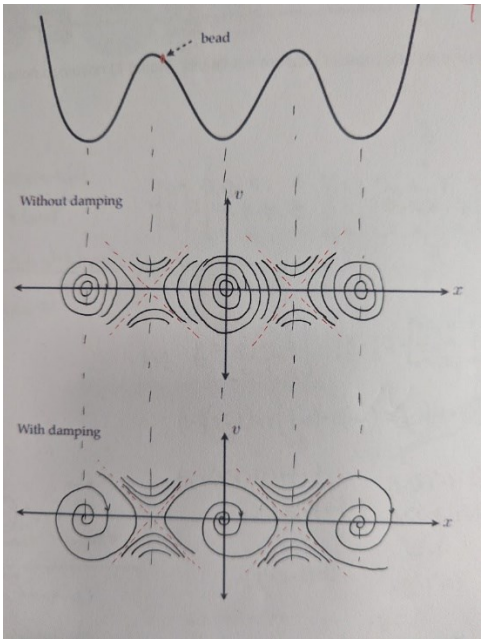
$$\vec{F} = -\nabla U = -\left(\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right)$$

$$\frac{\partial U}{\partial x} = \frac{\partial}{\partial x}[Axy^2 + B\sin(Cz)] = Ay^2$$
$$\frac{\partial U}{\partial y} = \frac{\partial}{\partial y}[Axy^2 + B\sin(Cz)] = 2Axy$$
$$\frac{\partial U}{\partial z} = \frac{\partial}{\partial z}[Axy^2 + B\sin(Cz)] = BC\cos(Cz)$$

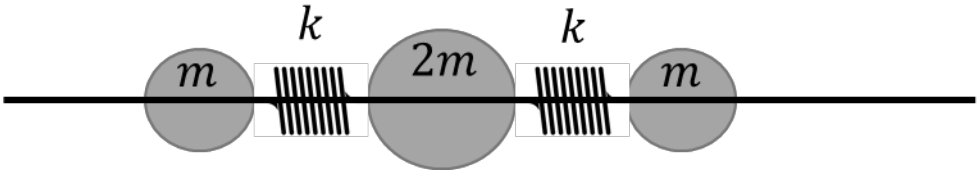
$$\vec{F} = -Ay^2\hat{i} - 2Axy\hat{j} - BC\cos(Cz)\hat{k}$$ or $\underline{F} = \begin{bmatrix} \\ \end{bmatrix}$

Q7 Consider the following physical system of a bead constrained to move the potential field (U(x)). But there are no equations given. Sketch the phase portrait ($x\ vs\ \dot{x}$) without or with damping. Starting with fixed points and think about their stability.

Graph made by Daniel:



Q8 Triatomic system: three masses move only in one direction and connected with two springs. To make our calculation easier let's do m=k=1.



Derive the equation of motion (1 token), find all the modes (1 token) and their directions (1 token).

Doing Newtonian mech here is ok:

$$m_1\ddot{x}_1 = k(x_2 - x_1) \implies \ddot{x}_1 + x_1 - x_2 = 0$$
$$m_2\ddot{x}_2 = -k(x_2 - x_1) + k(x_3 - x_2) \implies 2\ddot{x}_2 - x_1 + 2x_2 - x_3 = 0$$
$$m_3\ddot{x}_3 = -k(x_3 - x_2) \implies \ddot{x}_3 - x_2 + x_3 = 0$$

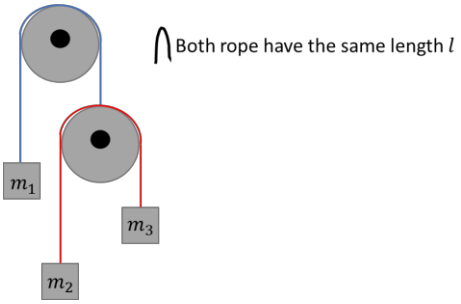
$$\det \begin{bmatrix} 1-\lambda & -1 & 0 \\ -1 & 2-2\lambda & -1 \\ 0 & -1 & 1-\lambda \end{bmatrix} = 0$$

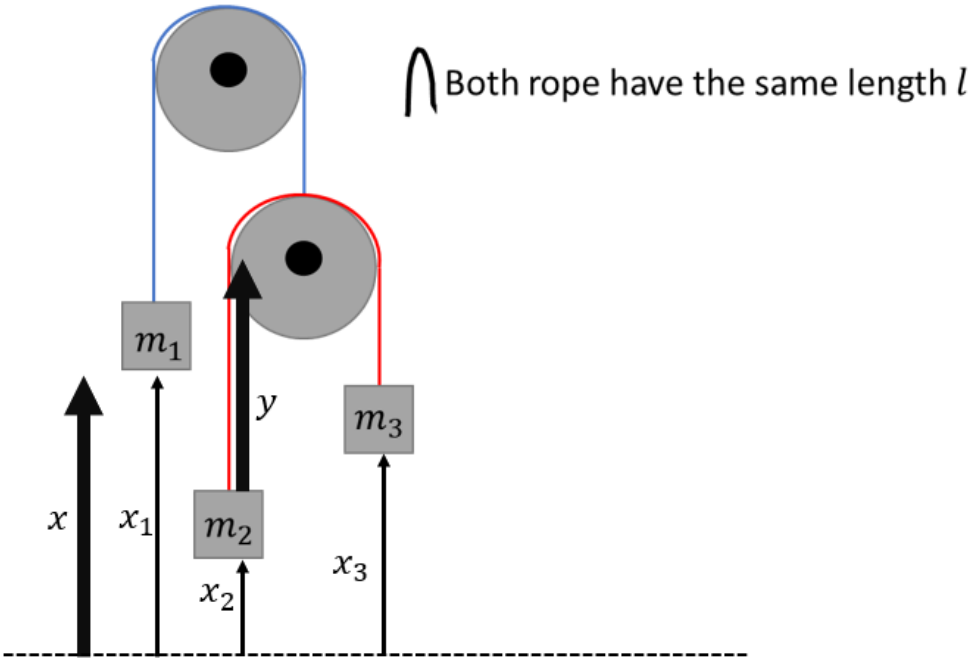
$$(1-\lambda)\left[(2-2\lambda)(1-\lambda) - 1\right] - (-1)\left[-(1-\lambda)\right] = 0$$
$$(1-\lambda)\left[2(1-\lambda)^2 - 2\right] = 2(1-\lambda)\left[(1-\lambda)^2 - 1\right] = 0$$
$$2(1-\lambda)(-\lambda)(2-\lambda) = 0$$

$$\lambda_1 = 0, \lambda_2 = 1, \lambda_3 = 2$$

$$\vec{a}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \vec{a}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad \vec{a}_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Q9 Double Atwood machine: derive the equation of motion of this. To simplify, assume both pulleys are mass-less (no moment of inertia), you can also ignore the radii of the pulleys. The length of the ropes (indicated with blue and red) are both l, and they are not elastic. Hint: 2 DOF.





$$x_1 = x \quad \dot{x}_1 = \dot{x}$$

$$x_2 = l - x + y \quad \dot{x}_2 = -\dot{x} + \dot{y}$$

$$x_3 = 2l - x - y \quad \dot{x}_3 = -\dot{x} - \dot{y}$$

$$
\begin{aligned}
T &= \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 + \frac{1}{2}m_3\dot{x}_3^2 \\
&= \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2(-\dot{x}+\dot{y})^2 + \frac{1}{2}m_3(-\dot{x}-\dot{y})^2 \\
&= \frac{1}{2}(m_1+m_2+m_3)\dot{x}^2 + \frac{1}{2}(m_2+m_3)\dot{y}^2 + (m_3-m_2)\dot{x}\dot{y}
\end{aligned}
$$

$$
\begin{aligned}
V &= -g[m_1x_1 + m_2x_2 + m_3x_3] \\
&= -g[m_1x + m_2(l-x+y) + m_3(2l-x-y)]
\end{aligned}
$$

$$
\frac{\partial L}{\partial \dot{x}} = (m_1+m_2+m_3)\dot{x} + (m_3-m_2)\dot{y}, \quad \frac{\partial L}{\partial x} = g(m_1-m_2-m_3)
$$
$$
\frac{\partial L}{\partial \dot{y}} = (m_3-m_2)\dot{x} + (m_2+m_3)\dot{y}, \quad \frac{\partial L}{\partial y} = g(m_2-m_3)
$$

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0$:

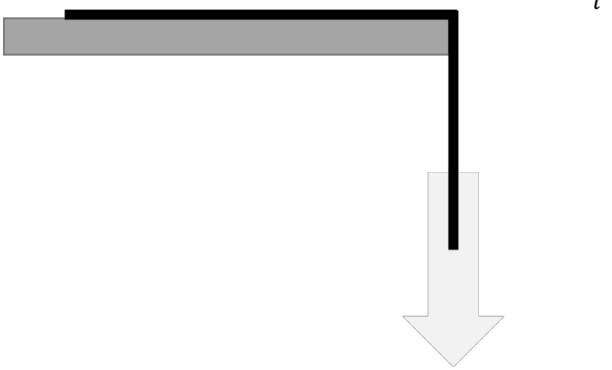
$$
\begin{aligned}
(m_1+m_2+m_3)\ddot{x} + (m_3-m_2)\ddot{y} &= g(m_1-m_2-m_3) \\
(m_3-m_2)\ddot{x} + (m_2+m_3)\ddot{y} &= g(m_2-m_3)
\end{aligned}
$$

Q10 Sliding… rope problem: Assuming *no friction*. A rope is sliding down the edge of a table. Derive the equation of motion using Lagrangian mech. (1 token) Then try to find, is the system stable? (1 token)

Total length of this rope is l

Total mass of this rope is m

So unit mass per length is $\frac{m}{l}$



$$L = \frac{1}{2}m\dot{x}^2 + \frac{mg}{2l}x^2$$

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x} \implies \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = m\ddot{x}, \quad \frac{\partial L}{\partial x} = \frac{mg}{l}x$$

$$m\ddot{x} - \frac{mg}{l}x = 0 \implies \ddot{x} - \frac{g}{l}x = 0$$

Eigenvalue it, will have [redacted] one positive real part, thus unstable.